■ $\exists x \in \mathbb{R}, x^3 - 2x + x = 0$ (witness: $x = 0$)

2. Disprove.

$\forall x \in \mathbb{Z}, \frac{x}{2} \in F$

Assume $x = 5$
$\frac{5}{2} = 2.5$
$2.5 \notin \mathbb{Z}$

■ $\exists x \in \mathbb{Z}, \frac{x}{2} \notin \mathbb{Z}$ (witness: $x = 5$)

3. Prove These Directly from the new definitions provided.

Definition A: f(x) is increasing on domain R if and only if
$\forall x \in \mathbb{R}, \forall k > 0, f(x) < f(x + k)$

Show that the sum of increasing functions is increasing

Assume f(x) is increasing.

Applying the definition of an increasing function:
$\forall x \in \mathbb{R}, \forall k > 0, f(x) < f(x + k)$
Assume the same for the function $g(x)$ and apply the same definition:
$\forall x \in \mathbb{R}, \forall k > 0, g(x) < g(x + k)$

By the nature of addition
$f(x) + g(x) < f(x + k) + g(x + k)\ \forall k > 0$

■ $\forall x \in \mathbb{R}, \forall k > 0, f(x) + g(x) < f(x + k) + g(x + k)$

Definition B: f(x) is p-periodic on R if and only if
$\forall x \in \mathbb{R}, f(x + p) = f(x)$

Assume $f(x)$ is perdiodic, $\forall x \in \mathbb{R}, f(x + p) = f(x)$
Assume $g(x)$ is perdiodic, $\forall x \in \mathbb{R}, g(x + p) = g(x)$
Let $h(x) = f(x) + g(x)$
Then $h(x + p) = f(x + p) + g(x + p)$

Therefore $h(x + p) = h(x)$
Or
$f(x + p) + g(x + p) = f(x) + g(x)$
■ which is p-peridiodic by the established definition.

Definition C: f(x) is bounded above if and only if
$\exists K \in \mathbb{R}, \forall x \in \mathbb{R}, f(x) \leq K$

Prove: if () is bounded above then 2() is bounded above

Let $2K = L$ and $2f(x) = g(x)$
It follows that $\exists L \in \mathbb{R}, \forall x \in \mathbb{R}, g(x) \leq L$
■ Therefore, $2f(x)$ $[g(x)]$ is bounded above by $2K$ (L)